

Solution of Exercise 5

Exercise 5.1 Singular case

- a) • Hamiltonian function $H(x, \lambda, u) = (x - y)^2 - \lambda ux$
• Optimality conditions

$$\dot{x} = \frac{\partial H}{\partial \lambda} = -ux, \quad x(0) = 0.1$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -2(x - y) + \lambda u, \quad \lambda(T) = 0$$

$$0 = \frac{\partial H}{\partial u} = -\lambda x \Rightarrow \text{independent of } u \Rightarrow \text{singular case}$$

- Consideration of the singular case: Search for $\frac{\partial}{\partial u} \left(\frac{d^r}{dt^r} H_u(\cdot) \right) \neq 0$, i.e. the first temporal derivative of $H_u = \frac{\partial H}{\partial u}$ in which the control input u appears explicitly.

$$\begin{aligned} \frac{d}{dt} H_u(\cdot) &= \frac{d}{dt} (-\lambda x) = -\dot{\lambda} x - \lambda \dot{x} \\ &= (2(x - y) - \lambda u)x + ux\lambda = 2x(x - y) - \lambda ux + \lambda ux \\ &= 2x(x - y) \end{aligned}$$

$$\begin{aligned} \frac{d^2}{dt^2} H_u(\cdot) &= \frac{d}{dt} (2x(x - y)) = 2\dot{x}(x - y) + 2x(\dot{x} - \dot{y}) \\ &= 2(-ux)(x - y) + 2x(-ux - \dot{y}) \\ &= -2ux^2 + 2uxy - 2ux^2 - 2x\dot{y} = 2xu(-2x + y) - 2x\dot{y} \stackrel{!}{=} 0 \\ &\Rightarrow u(-2x + y) - \dot{y} \stackrel{!}{=} 0 \\ &\Rightarrow u = \frac{\dot{y}}{y - 2x} \Rightarrow \text{degree of singularity } r = 2 \end{aligned}$$

- Consideration of the input constraints (Pontryagin's maximum principle)

$$\begin{aligned} u^* &= \arg \min_{u \in [u_{\min}, u_{\max}]} H = \arg \min_{u \in [u_{\min}, u_{\max}]} -\lambda ux \\ \Rightarrow u^* &= \begin{cases} u_{\min}, & \text{if } -\lambda x > 0 \quad || \quad (-\lambda x = 0 \quad \&\& \quad u^0 < u_{\min}) \\ u_{\max}, & \text{if } -\lambda x < 0 \quad || \quad (-\lambda x = 0 \quad \&\& \quad u^0 > u_{\max}) \\ u^0, & \text{if } -\lambda x = 0 \quad \text{with} \quad u^0 = \frac{\dot{y}}{y - 2x} \end{cases} \end{aligned}$$

b) Solution with the regularization term

- Hamiltonian function $H(x, \lambda, u) = (x - y)^2 + \frac{c}{2} u^2 - \lambda ux$
• Optimality conditions

$$\dot{x} = \frac{\partial H}{\partial \lambda} = -ux, \quad x(0) = 0.1$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = -2(x - y) + \lambda u, \quad \lambda(T) = 0$$

$$0 = \frac{\partial H}{\partial u} = cu - \lambda x \Rightarrow u^0 = \frac{\lambda x}{c}$$

- Consideration of the input constraints (Pontryagin's maximum principle)

$$u^* = \arg \min_{u \in [u_{\min}, u_{\max}]} H = \arg \min_{u \in [u_{\min}, u_{\max}]} \frac{c}{2} u^2 - \lambda u x$$

$$\Rightarrow u^* = \begin{cases} u_{\min}, & \text{if } u^0 \leq u_{\min} \\ u^0, & \text{if } u^0 \in (u_{\min}, u_{\max}) \\ u_{\max}, & \text{if } u^0 \geq u_{\max} \end{cases}$$

- Plots for different parameter values:

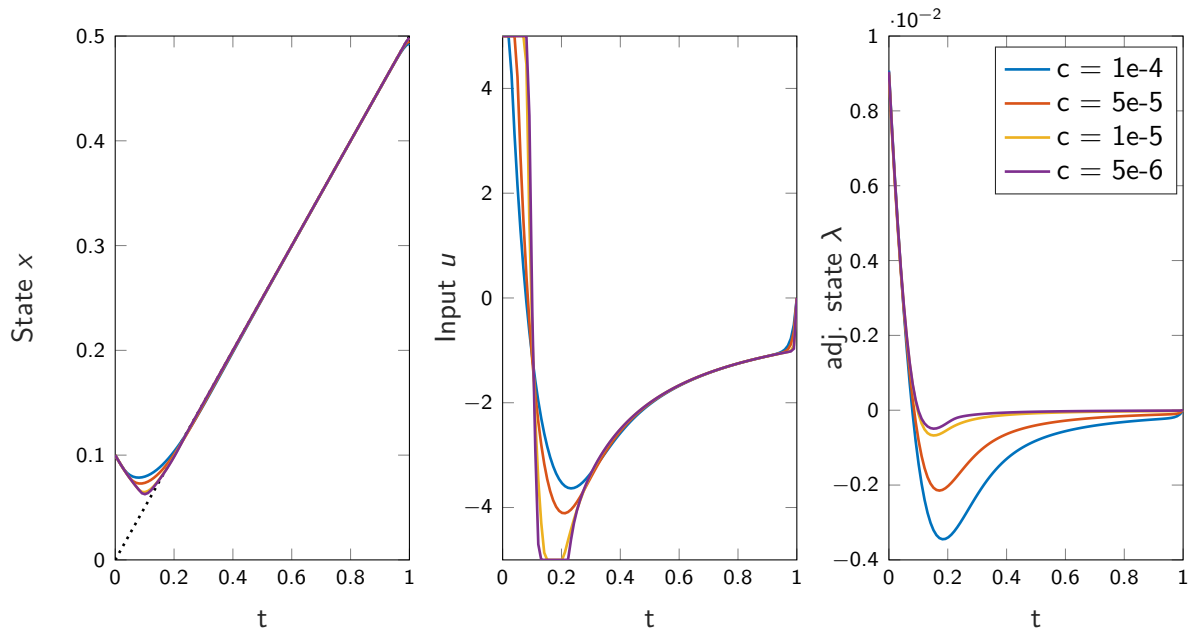


Figure 1: Trajectories for different parameter values for $x(0) = 0.1$.

Exercise 5.2 Linear quadratic regulator (LQR)

- a) • State-space form of the system dynamics

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{g}{l} \sin(x_1) - \frac{b}{ml^2} \omega + \frac{1}{ml^2} u \end{bmatrix}.$$

- Linearization around $(\bar{\mathbf{x}}, \bar{u})$ with $\bar{\mathbf{x}} = [\pi/2, 0]^T$, $\bar{u} = 10$

$$\Delta \dot{\mathbf{x}} = \mathbf{A} \Delta \mathbf{x}(t) + \mathbf{B} \Delta u(t), \quad \text{with}$$

$$\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{(\bar{\mathbf{x}}, \bar{u})} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} \cos(\bar{x}_1) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix},$$

$$\mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial u} \right|_{(\bar{\mathbf{x}}, \bar{u})} = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

- b) • Standard LQR form

$$\min_{\Delta u(\cdot)} \int_0^\infty \left(\Delta \mathbf{x}(t)^T \mathbf{Q} \Delta \mathbf{x}(t) + \Delta u(t)^T \mathbf{R} \Delta u(t) \right) dt$$

$$\text{s.t. } \Delta \dot{\mathbf{x}}(t) = \mathbf{A} \Delta \mathbf{x}(t) + \mathbf{B} \Delta u(t)$$

$$\text{with } \mathbf{Q} = \beta^2 \mathbf{I} \in \mathbb{R}^{2 \times 2}, \quad \mathbf{R} = \alpha^2.$$

- The optimal feedback law is

$$\Delta u^*(t) = -\mathbf{K}\Delta \mathbf{x}(t), \quad \text{with} \quad \mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P},$$

where $\mathbf{P}$ is the positive semidefinite solution of the continuous algebraic Riccati equation

$$\mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} + \mathbf{Q} = 0.$$

- c) • For $\alpha = 1$ and $\beta = 2$ follows $\mathbf{Q} = 4\mathbf{I}$, $\mathbf{R} = 1$.
- Let $\mathbf{P} = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$, then the algebraic Riccati equation can be written as

$$\begin{aligned} \begin{bmatrix} 0 & 0 \\ p_1 - p_2 & p_2 - p_3 \end{bmatrix} + \begin{bmatrix} 0 & p_1 - p_2 \\ 0 & p_2 - p_3 \end{bmatrix} - \begin{bmatrix} 2p_2 \\ 2p_3 \end{bmatrix} \begin{bmatrix} 2p_2 & 2p_3 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} &= 0 \\ \begin{bmatrix} 0 & p_1 - p_2 \\ p_1 - p_2 & 2p_2 - 2p_3 \end{bmatrix} - \begin{bmatrix} 4p_2^2 & 4p_2p_3 \\ 4p_2p_3 & 4p_3^2 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} &= 0. \end{aligned}$$

Therefore

$$\begin{aligned} -4p_2^2 + 4 &= 0 \Rightarrow p_2 = 1 \\ p_1 - p_2 - 4p_2p_3 &= 0 \\ 2p_2 - 2p_3 - 4p_3^2 + 4 &= 0 \xrightarrow{p_2=1} 4p_3^2 + 2p_3 - 6 = 0 \\ &\Rightarrow \underbrace{(2p_3 + 3)}_{p_3 = -\frac{3}{2}} \underbrace{(p_3 - 1)}_{p_3 = 1} = 0 \Rightarrow p_3 = 1 \quad (P \succeq 0). \end{aligned}$$

With $p_2 = 1$ and $p_3 = 1$ follows

$$p_1 - p_2 - 4p_2p_3 = 0 \Rightarrow p_1 = 5.$$

- The feedback gain is

$$\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \end{bmatrix}.$$

- The closed-loop matrix is

$$\mathbf{A}_{cl} = \mathbf{A} - \mathbf{B}\mathbf{K} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix},$$

and its eigenvalues are

$$\begin{aligned} \det(s\mathbf{I} - \mathbf{A}_{cl}) &\stackrel{!}{=} 0 \\ \det\left(\begin{bmatrix} s & -1 \\ 4 & s+5 \end{bmatrix}\right) &= s(s+5) - (-1) \cdot 4 \\ &= s^2 + 5s + 4 \Rightarrow \boldsymbol{\lambda} = \{-1, -4\}. \end{aligned}$$

Exercise 5.3 Fishing problem

- a) • The fish population is stationary, if

$$\dot{x} = 0 \Rightarrow 0 = bx(t) \left(1 - \frac{x(t)}{c}\right) - u(t)x(t).$$

- For $u = 0$:

$$0 = bx(t) \left(1 - \frac{x(t)}{c}\right) \Rightarrow x(t) = c$$

Without fishing, the fish population reaches the capacity limits of the lake.

- For $u = b$:

$$0 = bx(t) \left(1 - \frac{x(t)}{c}\right) - bx(t) = -\frac{b}{c}x(t)^2 \Rightarrow x(t) = 0$$

For maximum fishing, all fish are harvested and the population dies out.

- b) • The minimization problem is

$$\begin{aligned} \min_{u(\cdot)} \quad & \int_0^{t_f} -pxu \, dt \\ \text{s.t.} \quad & \dot{x} = bx \left(1 - \frac{x}{c}\right) - ux, \quad x(0) = x_0 \\ & u \in [0, b]. \end{aligned}$$

- Hamiltonian function $H(x, \lambda, u) = -pxu + \lambda \left(bx \left(1 - \frac{x}{c}\right) - ux\right)$
 • Optimality conditions

$$\begin{aligned} \dot{x} &= \frac{\partial H}{\partial \lambda} = bx \left(1 - \frac{x}{c}\right) - ux, \quad x(0) = x_0 \\ \dot{\lambda} &= -\frac{\partial H}{\partial x} = pu - \lambda \left(b - \frac{2b}{c}x - u\right), \quad \lambda(t_f) = 0. \end{aligned}$$

- Consideration of the input constraints (Pontryagin's maximum principle)

$$\begin{aligned} u^*(\lambda) &= \arg \min_{u \in [0, b]} H = \arg \min_{u \in [0, b]} [-xu(p + \lambda)] \\ \Rightarrow u^*(\lambda) &= \begin{cases} 0, & \text{if } p + \lambda > 0, \\ b, & \text{if } p + \lambda < 0. \end{cases} \end{aligned}$$

- $\lambda(t)$ shows the sensitivity of the harvest at time t with respect to the fish population $x(t)$.
 – If $\lambda(t) > -p$, then the fish is more valuable in the lake than being sold, hence $u = 0$.
 – If $\lambda(t) < -p$, then selling the fish is more valuable than its future benefit, hence $u = b$.

- c) • Singular case

$$\begin{aligned}\frac{\partial H}{\partial u} = 0 &\Rightarrow 0 = -px - \lambda x = -(p + \lambda)x \Rightarrow \lambda = -p = \text{const.} \\ \frac{d}{dt}H_u(\cdot) = 0 &\Rightarrow 0 = -(p + \lambda)\dot{x} - \dot{\lambda}x \Rightarrow \dot{\lambda} = 0 \\ &\Rightarrow \dot{\lambda} = pu - b\lambda + \frac{2b}{c}\lambda x + \lambda u \stackrel{!}{=} 0 \\ &\Rightarrow x_s = \frac{c}{2}.\end{aligned}$$

- Corresponding control

$$\begin{aligned}\dot{x} &= bx_s \left(1 - \frac{x_s}{c}\right) - u_s x_s \stackrel{!}{=} 0 \\ \Rightarrow b \frac{c}{2} \left(1 - \frac{1}{2}\right) - u_s \frac{c}{2} &= 0 \\ \Rightarrow u_s &= \frac{b}{2}.\end{aligned}$$

- d) • The natural growth of the population is given by

$$g(x) = bx \left(1 - \frac{x}{c}\right).$$

- Since

$$g'(x) = b - \frac{2b}{c}x \stackrel{!}{=} 0 \Rightarrow x = \frac{c}{2},$$

the singular fish population $x_s = \frac{c}{2}$ maximizes the natural growth rate of the fish population.

Add-On 1 Maximum Principle for Time-Invariant Systems

- a) • Equivalent minimization problem

$$\begin{aligned} \min_{u(\cdot)} \quad & -x(t_f) - \int_0^{t_f} (1 - u(t))x(t) dt \\ \text{s.t.} \quad & \dot{x}(t) = \alpha u(t)x(t), \quad x(0) = x_0 \\ & u(t) \in [0, 1], \quad \forall t \geq 0. \end{aligned}$$

- Hamiltonian function $H(x, \lambda, u) = -(1 - u)x + \lambda \alpha u x$
- Adjoint dynamics

$$\begin{aligned} \dot{x} &= \frac{\partial H}{\partial \lambda} = \alpha u x, \quad x(0) = x_0 \\ \dot{\lambda} &= -\frac{\partial H}{\partial x} = 1 - u - \alpha \lambda u, \quad \lambda(t_f) = V_x(t_f) = -1 \end{aligned}$$

- b) • Consideration of the input constraints (Pontryagin's maximum principle)

$$\begin{aligned} u^*(\lambda) &= \arg \min_{u \in [0,1]} H = \arg \min_{u \in [0,1]} [x u (1 + \alpha \lambda)] \\ &= \begin{cases} 1, & \text{if } 1 + \alpha \lambda(t) < 0, \\ 0, & \text{if } 1 + \alpha \lambda(t) > 0, \\ \tilde{u} \in [0, 1], & \text{if } 1 + \alpha \lambda(t) = 0, \end{cases} \quad \text{for } x(t) > 0. \end{aligned}$$

- The switching function is $\sigma(t) = 1 + \alpha \lambda(t)$.
- c) • From $\lambda(t_f) = -1$ follows

$$\begin{aligned} \sigma(t_f) &= 1 + \alpha \lambda(t_f) = 1 - \alpha > 0 \\ u^*(t_f) &= 0. \end{aligned}$$

Hence, the final interval must satisfy

$$u^*(t) = 0, \quad t' \leq t \leq t_f,$$

and therefore

$$u^*(t) = \begin{cases} 1, & 0 \leq t \leq t', \\ 0, & t' \leq t \leq t_f. \end{cases}$$

- Determine the switching time

$$\begin{aligned} u^*(t) = 0 &\Rightarrow \dot{\lambda}(t) = 1 \Rightarrow \lambda(t) = t + C \\ \lambda(t_f) = -1 &\Rightarrow C = -t_f - 1 \Rightarrow \lambda(t) = t - t_f - 1. \end{aligned}$$

With the switching function at the switching time t' follows

$$\begin{aligned} \sigma(t') = 0 &\Rightarrow 1 + \alpha \lambda(t') = 0 \Rightarrow \lambda(t') = -\frac{1}{\alpha} \\ &\Rightarrow t' - t_f - 1 = -\frac{1}{\alpha} \\ &\Rightarrow t' = t_f + 1 - \frac{1}{\alpha}. \end{aligned}$$